

PASS: Predictive Auto-Scaling System in Cloud Environments

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Motivation for Predictive Auto-Scaling

Challenges with Reactive Auto-Scaling

- Reactive scaling adds instances after load increases which causes delay.
- Instances take time to start(warm-up time), so system performance drops temporarily.
- Sudden traffic spikes can reduce the QoS (Quality of Service).
- Reactive scaling often leads to either over-provisioning (high cost) or under-provisioning (slow response).

Solutions provided by Predictive Scaling

- Forecasts workload in advance using past traffic patterns.
- Scales resources before demand increases, removing startup delay.
- Avoids performance drops and maintains QoS during sudden spikes.
- Balances cost and performance — adds only the instances that are actually needed.

Selection of Predictive Scaling Algorithm(ELPA)

Online Models

- Continuously learn and update with real-time, recent data.
- Quickly adapt to short-term changes and normal workload fluctuations.

Challenges with using only Online model

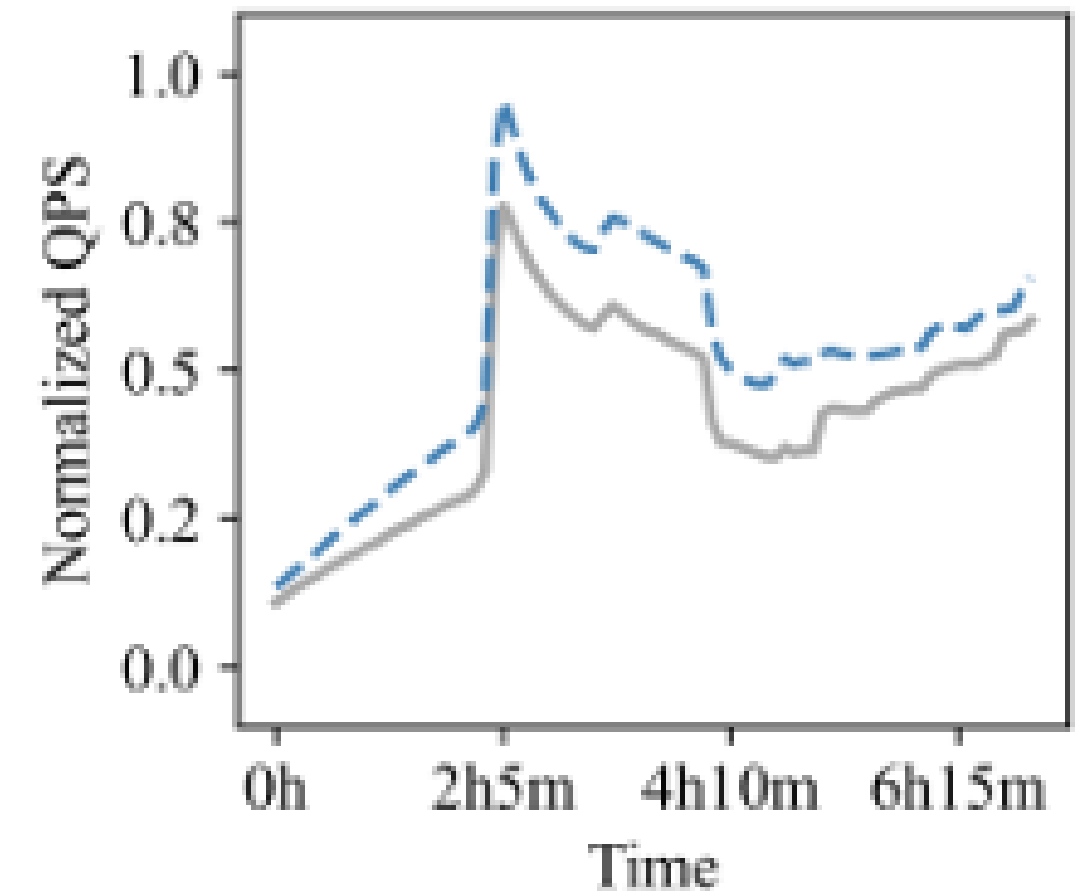
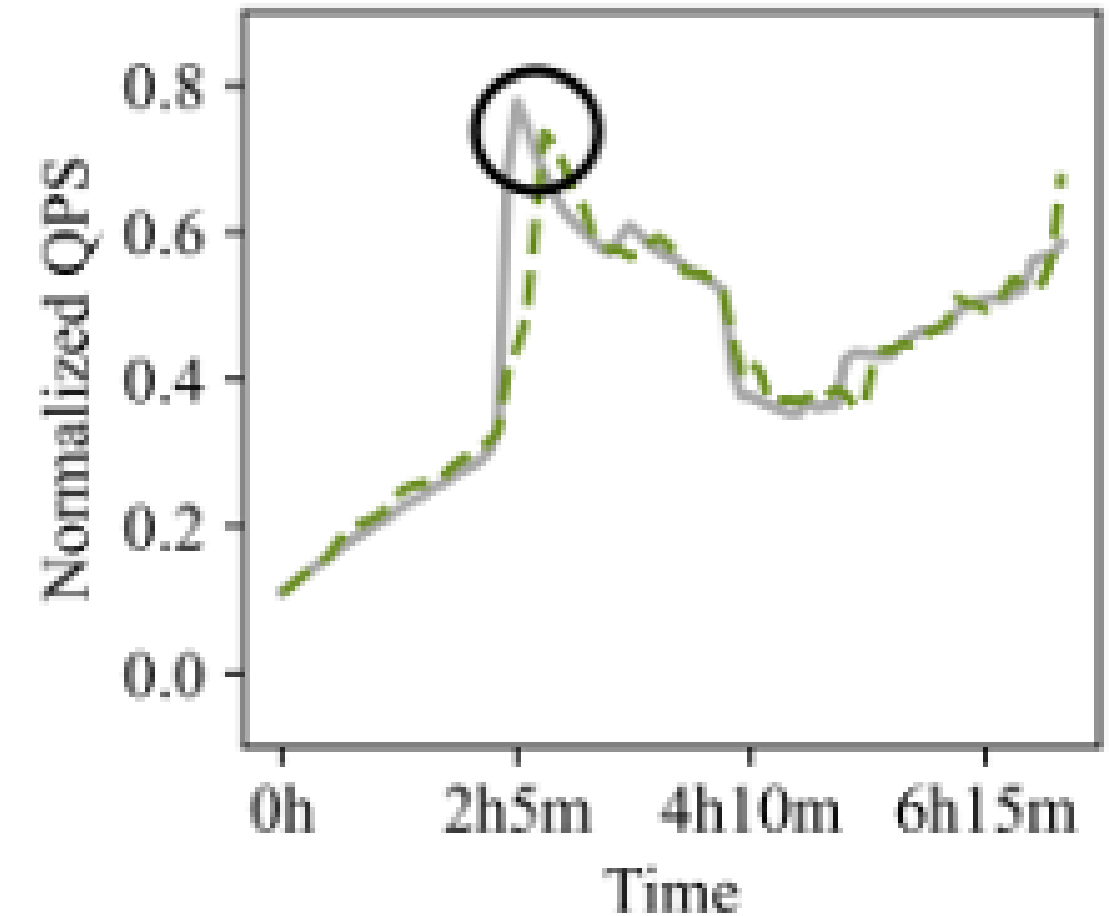
- There might be a delay if the workload suddenly changes.
- They lag behind sudden spikes or drops, causing inaccurate predictions.

Offline Models

- Trained using historical data and capture long-term periodic patterns.
- Handle large-scale or sudden variations better but don't update continuously.

Challenges with using only Offline model

- Their predictions may have amplitude errors (over/underestimating traffic size).
- The workload pattern is smooth and stable.



How ELPA works

Runs two kinds of predictors:

- Online model: uses recent data and adapts quickly to short-term, stable trends.
- Offline model: uses long-term historical data to capture periodic or sudden large changes.
- Selects between them using recent errors caused by both models

Performance Model

The Performance Model maps the predicted workload (QPS) to the number of instances needed

The model stores, for each instance count, the maximum QPS that can be handled.

Creation process





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Mind map

Exploring creativity

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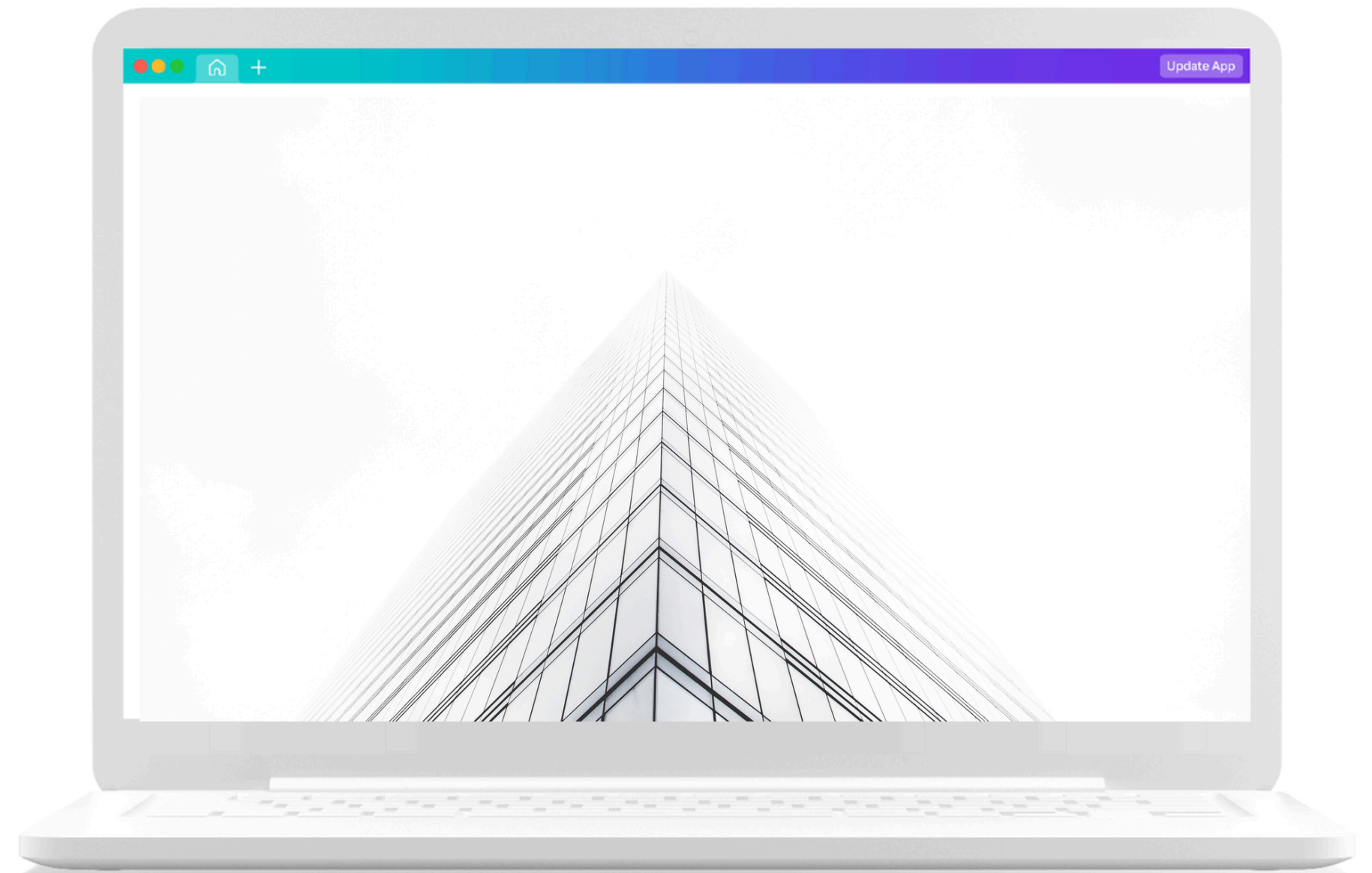
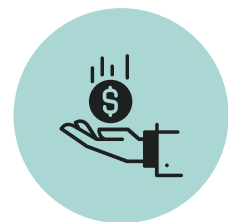


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Mockups

Exploring creativity

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Strengths

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Weaknesses

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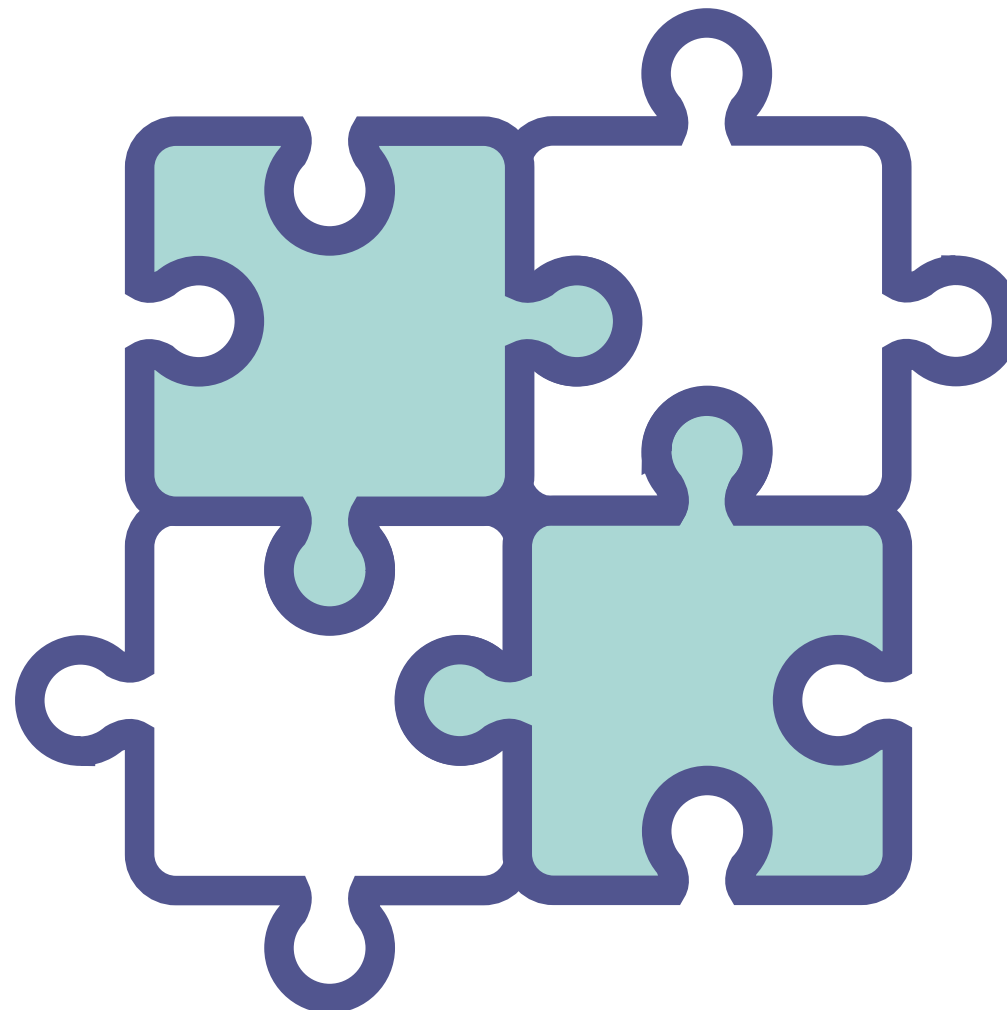
Threats

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Opportunities

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**Thank you
very much!**

PRESENTED BY HELENE PAQUET